

Production-Level LMS (Learning Management System) Guide

1. Core Modules

- 1 Student: Authentication, course browsing, video player, progress tracking, notes, quizzes, certificates
- 2 Instructor: Course creation, video uploads, student management, earnings dashboard
- 3 Admin: User management, course approvals, analytics, system control

2. Payment & Monetization

- 1 Razorpay or Stripe integration
- 2 Course purchase system
- 3 Subscriptions (optional)
- 4 Coupons and discounts
- 5 Invoice generation

3. Advanced Video Features

- 1 HLS streaming
- 2 Resume playback
- 3 Playback speed control
- 4 Auto-next lesson
- 5 Watermark protection

4. Learning Features

- 1 Quizzes with timers
- 2 Assignments
- 3 Auto grading
- 4 Discussion system
- 5 Live classes

5. Analytics

- 1 Student progress tracking
- 2 Course performance metrics
- 3 Drop-off analysis
- 4 Instructor earnings insights
- 5 Admin dashboards

6. Engagement System

- 1 Email notifications
- 2 In-app notifications
- 3 Reminders
- 4 Deadline alerts

7. Security

- 1 Role-based authentication
- 2 Protected routes
- 3 Video protection
- 4 Rate limiting
- 5 Input validation

8. UI/UX

- 1 Clean and professional design
- 2 Minimal animations
- 3 Responsive layout
- 4 Good typography and spacing
- 5 Dark/light mode

9. Tech Stack

- 1 Frontend: React, TypeScript, Tailwind
- 2 Backend: Node.js, Express
- 3 Database: PostgreSQL or MongoDB
- 4 Storage: AWS S3 or Cloudinary
- 5 Authentication: JWT or Cookies

10. Bonus Features

- 1 AI recommendations
- 2 Chat system
- 3 Gamification
- 4 Multi-language support
- 5 Offline downloads
- 6 Admin CMS